

Name=Value in Function Calls

Since R2021a

MATLAB® supports two syntaxes for passing name-value arguments.

`plot(x,y,LineWidth=2)` **name=value syntax**

`plot(x,y,"LineWidth",2)` **comma-separated syntax**

Use the name=value syntax to help identify name-value arguments for functions and to clearly distinguish names from values in lists of name-value arguments.

Most functions and methods support both syntaxes, but there are some limitations on where and how the name=value syntax can be used:

- **Mixing name, value and name=value syntaxes:** The recommended practice is to use only one syntax in any given function call. However, if you do mix name=value and name, value syntaxes in a single call, all name=value arguments must appear after the name, value arguments. For example, `plot(x,y,"Color","red",LineWidth=2)` is a valid combination, but `plot(x,y,Color="red","LineWidth",2)` errors.
- **Using positional arguments after name-value arguments:** Some functions have positional arguments that appear after name-value arguments. For example, this call to the `verifyEqual` method uses the `RelTol` name-value argument, followed by a string input:

```
verifyEqual(testCase,1.5,2,"RelTol",0.1,...  
    "Difference exceeds relative tolerance.")
```

Using the name=value syntax (`RelTol=0.1`) causes the statement to error. In cases where a positional argument follows name-value arguments, use the name, value syntax.

- **Names that are invalid variable names:** Name-value arguments with names that are invalid MATLAB variable names cannot be used with the name=value syntax. See [Variable Names](#) for more info. For example, a name-value argument like `"allow-empty",true` errors if passed as `allow-empty=true`. Use the name, value syntax in these cases.

Function authors do not need to code differently to support both the name, value and name=value syntaxes. For information on using argument validation with name-value arguments, see [Validate Name-Value Arguments](#).

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